

Introduction to the Density Matrix Renormalization Group (DMRG)

Tensor Network Methods Lecture

1 Motivation

Many-body quantum systems have Hilbert spaces that grow exponentially with system size. For example, a spin- $\frac{1}{2}$ chain of length L has

$$\dim(\mathcal{H}) = 2^L \tag{1}$$

Exact diagonalization therefore becomes impossible for large L . Several renormalization approaches attempt to overcome this problem:

- Block Renormalization Group (BRG)
- Numerical Renormalization Group (NRG)
- Density Matrix Renormalization Group (DMRG)

DMRG, introduced by Steven White in 1992, is today the most powerful numerical method for one-dimensional quantum systems [1]. The key idea is:

Instead of keeping the lowest-energy states of a block (as in BRG), keep the states that are most important in the *ground state wavefunction*.

This importance is determined using the **reduced density matrix**.

2 Density Matrix Formalism

2.1 Pure States

A pure quantum state is described by a wavefunction

$$|\psi\rangle = \sum_i c_i |i\rangle \quad (2)$$

The basis can be the eigenvectors of the Hamiltonian. The corresponding density matrix is

$$\rho = |\psi\rangle \langle\psi| \quad (3)$$

Matrix elements:

$$\rho_{ij} = c_i c_j^* \quad (4)$$

Properties:

- $\rho^\dagger = \rho$
- $\text{Tr}(\rho) = 1$
- $\rho^2 = \rho$

2.2 Expectation Values and the Density Matrix

The expectation value of an operator A is by definition

$$\langle A \rangle = \langle\psi| A |\psi\rangle. \quad (5)$$

Expanding the state in the chosen basis, we obtain

$$\langle A \rangle = \left(\sum_i c_i^* \langle i| \right) A \left(\sum_j c_j |j\rangle \right) = \sum_{i,j} c_i^* c_j \langle i| A |j\rangle. \quad (6)$$

Now define the density matrix

$$\rho = |\psi\rangle \langle\psi|. \quad (7)$$

Its matrix elements are

$$\rho_{ji} = \langle j| \rho |i\rangle = c_j c_i^*. \quad (8)$$

Therefore we can rewrite the expectation value as

$$\langle A \rangle = \sum_{i,j} \rho_{ji} A_{ij} = \text{Tr}(\rho A) \quad (9)$$

2.3 Mixed States

More generally a system may be in a statistical mixture

$$\rho = \sum_n p_n |\psi_n\rangle \langle \psi_n| \quad (10)$$

with probabilities p_n . For mixed states

$$\rho^2 \neq \rho \quad (11)$$

$$\text{Tr}(\rho^2) < 1 \quad (12)$$

Important note: A pure quantum state can exist in a *superposition* of basis states,

$$|\psi\rangle = \sum_i c_i |i\rangle, \quad (13)$$

which means that the system simultaneously occupies all basis states with complex amplitudes c_i . This can lead to interference effects. In contrast, a *statistical mixture* describes classical uncertainty about which state the system is in. It is represented by a density matrix of the form

$$\rho = \sum_n p_n |\psi_n\rangle \langle \psi_n|, \quad (14)$$

where the system is in state $|\psi_n\rangle$ with probability p_n . Unlike a superposition, a statistical mixture cannot produce quantum interference. In other words, a superposition represents coherent quantum uncertainty, whereas a statistical mixture represents classical ignorance about the state of the system.

3 Reduced Density Matrix

Consider a system divided into two parts

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E \quad (15)$$

Where S is for the system and E for the environment. The ground state can be written as

$$|\psi\rangle = \sum_{i,j} \psi_{ij} |i\rangle_S |j\rangle_E \quad (16)$$

The reduced density matrix of the system is obtained by tracing out the environment

$$\rho_S = \text{Tr}_E |\psi\rangle \langle \psi| \quad (17)$$

Matrix elements

$$(\rho_S)_{ii'} = \sum_j \psi_{ij} \psi_{i'j}^* \quad (18)$$

This object contains all information about measurements performed on the subsystem. For an observable A acting only on the system

$$\langle A \rangle = \text{Tr}_S(\rho_S A) \quad (19)$$

This will be crucial for DMRG.

3.1 Entropy

An important quantity associated with the reduced density matrix is the *von Neumann entropy*. For a subsystem described by the reduced density matrix ρ_S , the entropy is defined as

$$S(\rho_S) = -\text{Tr}(\rho_S \log \rho_S). \quad (20)$$

If ρ_S is diagonalized as

$$\rho_S = \sum_{\alpha} w_{\alpha} |w_{\alpha}\rangle \langle w_{\alpha}|, \quad (21)$$

the entropy can be written as

$$S = -\sum_{\alpha} w_{\alpha} \log w_{\alpha}. \quad (22)$$

Properties The von Neumann entropy has several important properties:

- $S = 0$ if and only if the subsystem is in a pure state.
- $S > 0$ if the subsystem is entangled with the rest of the system.
- For a pure bipartite state, the entropies of the two subsystems are equal:

$$S(\rho_S) = S(\rho_E). \quad (23)$$

- The entropy is maximal for a maximally mixed state.

The entropy is bounded from above by the dimension M_S of the Hilbert space of the subsystem. The maximal value occurs for a maximally mixed state. In this case all eigenvalues of the density matrix are equal,

$$w_{\alpha} = \frac{1}{M_S}, \quad (24)$$

and the entropy becomes

$$S_{\max} = -\sum_{\alpha=1}^{M_S} \frac{1}{M_S} \log \frac{1}{M_S} = \log M_S. \quad (25)$$

Thus the entropy is bounded as

$$0 \leq S(\rho_S) \leq \log M_S. \tag{26}$$

The entropy of the reduced density matrix is often called the *entanglement entropy*. It quantifies how strongly the subsystem is entangled with the rest of the system. In DMRG this quantity is closely related to the spectrum of the density matrix eigenvalues w_α . Rapid decay of these eigenvalues implies small entanglement entropy and allows an efficient truncation of the Hilbert space. For one-dimensional gapped systems the entanglement entropy typically satisfies an *area law*, meaning that it remains bounded as the system size increases. This property explains why the DMRG algorithm and matrix product states provide very accurate representations of ground states of one-dimensional local Hamiltonians.

4 Derivation of the Density Matrix Truncation Rule

We divide the Hilbert space into system and environment

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E \quad (27)$$

with orthonormal bases $\{|i\rangle_S\}$ and $\{|j\rangle_E\}$. The exact ground state can be written as

$$|\psi\rangle = \sum_i^{D_S} \sum_j^{D_E} \psi_{ij} |i\rangle_S |j\rangle_E. \quad (28)$$

Suppose the system Hilbert space has dimension D_S . We want to truncate it to dimension M_S by keeping only M_S orthonormal states $\{|\alpha\rangle\}_{\alpha=1}^{M_S}$. We approximate the state as

$$|\tilde{\psi}\rangle = \sum_{\alpha=1}^{M_S} \sum_j^{D_E} a_{\alpha j} |\alpha\rangle_S |j\rangle_E. \quad (29)$$

The goal is to choose the states $|\alpha\rangle$ such that the approximation is optimal. We therefore minimize

$$\| |\psi\rangle - |\tilde{\psi}\rangle \|^2. \quad (30)$$

Assuming real coefficients for simplicity and because $|\psi\rangle$ is normalized

$$\| |\psi\rangle - |\tilde{\psi}\rangle \|^2 = 1 - 2 \langle \tilde{\psi} | \psi \rangle + \sum_{\alpha}^{M_S} \sum_j^{D_E} |a_{\alpha j}|^2 \quad (31)$$

Write the truncated basis states as

$$|\alpha\rangle = \sum_i^{D_S} U_{\alpha i} |i\rangle_S. \quad (32)$$

Where U is non-square ($M_S \times D_S$) unitary matrix that rotates into a new basis. Then

$$|\tilde{\psi}\rangle = \sum_{\alpha}^{M_S} \sum_j^{D_E} a_{\alpha j} \sum_i^{D_S} U_{\alpha i} |i\rangle_S |j\rangle_E. \quad (33)$$

The overlap becomes

$$\langle \tilde{\psi} | \psi \rangle = \sum_{\alpha}^{M_S} \sum_j^{D_E} \sum_i^{D_S} a_{\alpha j} U_{\alpha i} \psi_{ij}. \quad (34)$$

Therefore

$$\| |\psi\rangle - |\tilde{\psi}\rangle \|^2 = 1 - 2 \sum_{\alpha}^{M_S} \sum_j^{D_E} \sum_i^{D_S} a_{\alpha j} U_{\alpha i} \psi_{ij} + \sum_{\alpha}^{M_S} \sum_j^{D_E} |a_{\alpha j}|^2. \quad (35)$$

For fixed $|\alpha\rangle$ we minimize with respect to $a_{\alpha j}$. The minimum is obtained for

$$a_{\alpha j} = \sum_i^{D_S} U_{\alpha i} \psi_{ij} \quad (36)$$

This means that the best approximation is obtained by projecting $|\psi\rangle$ onto the chosen subspace. Substituting back we obtain

$$\| |\psi\rangle - |\tilde{\psi}\rangle \|^2 = 1 - \sum_{\alpha}^{M_S} \sum_i^{D_S} \sum_{i'}^{D_S} U_{\alpha i} \left(\sum_j^{D_E} \psi_{ij} \psi_{i'j} \right) U_{\alpha i'} = 1 - \sum_{\alpha}^{M_S} \sum_i^{D_S} \sum_{i'}^{D_S} U_{\alpha i} \rho_{ii'} U_{\alpha i'} \quad (37)$$

The error becomes

$$\| |\psi\rangle - |\tilde{\psi}\rangle \|^2 = 1 - \sum_{\alpha=1}^{M_S} \langle \alpha | \rho_S | \alpha \rangle. \quad (38)$$

We must therefore choose M_S orthonormal states $|\alpha\rangle$ that maximize

$$\sum_{\alpha=1}^{M_S} \langle \alpha | \rho_S | \alpha \rangle. \quad (39)$$

Let the eigenvalues of ρ_S be

$$w_1 \geq w_2 \geq w_3 \geq \dots \quad (40)$$

with eigenvectors $|w_1\rangle, |w_2\rangle, \dots$. The maximum is obtained by choosing

$$|\alpha\rangle = |w_{\alpha}\rangle, \quad \alpha = 1, \dots, M_S. \quad (41)$$

Therefore the optimal truncated basis consists of the M_S eigenvectors of the reduced density matrix with the largest eigenvalues. With this choice

$$\sum_{\alpha=1}^{M_S} \langle \alpha | \rho_S | \alpha \rangle = \sum_{\alpha=1}^{M_S} w_{\alpha}. \quad (42)$$

The minimum error is therefore

$$\| |\psi\rangle - |\tilde{\psi}\rangle \|^2 = 1 - \sum_{\alpha=1}^{M_S} w_{\alpha}. \quad (43)$$

Defining the discarded weight

$$\epsilon = 1 - \sum_{\alpha=1}^{M_S} w_{\alpha} \quad (44)$$

we obtain the DMRG truncation rule:

$$\boxed{\text{Keep the eigenvectors of the reduced density matrix with the largest eigenvalues}} \quad (45)$$

and the truncation error is given by the discarded weight ϵ .

Failure of Block RG In Block RG we

1. Diagonalize a block Hamiltonian
2. Keep the M_S lowest-energy states
3. Build a larger block

This assumes the ground state of the full system is composed of low-energy states of each block. However this assumption is often incorrect. White and Noack showed that wrong boundary conditions inside blocks lead to large errors [1]. The solution:

Select states based on their contribution to the full system ground state.

This leads to DMRG.

Key Idea of DMRG Embed the system block in an environment and compute the ground state of the combined system. Then use the reduced density matrix of the block to select the most important states.

5 Infinite-System DMRG

We now describe the infinite-system DMRG algorithm in more detail. The system is divided into two blocks:

$$\text{System block} + \text{Environment block.} \quad (46)$$

The algorithm proceeds iteratively.

1. Start with a small system block of length ℓ with basis states $\{|m_\ell\rangle\}$.
2. Add one site with local basis $\{|\sigma_{\ell+1}\rangle\}$.
3. Construct the enlarged basis

$$|m_\ell \sigma_{\ell+1}\rangle = |m_\ell\rangle \otimes |\sigma_{\ell+1}\rangle. \quad (47)$$

4. Build the superblock Hamiltonian and compute the ground state

$$|\psi\rangle = \sum_{m_S}^{M_S} \sum_{m_E}^{M_E} \sum_{\sigma_S}^d \sum_{\sigma_E}^d \psi_{m_S \sigma_S \sigma_E m_E} |m_S \sigma_S\rangle |\sigma_E m_E\rangle. \quad (48)$$

5. Construct the reduced density matrix of the system (note: after reshaping the wavefunction it is only a matrix matrix multiplication

$$\rho_{(m_S \sigma_S), (m'_S \sigma'_S)} = \sum_{m_E}^{M_E} \sum_{\sigma_E}^d \psi_{m_S \sigma_S \sigma_E m_E} \psi_{m'_S \sigma'_S \sigma_E m_E}^*. \quad (49)$$

6. Diagonalize ρ and keep the M_S eigenvectors with the largest eigenvalues.
7. These eigenvectors define the new truncated basis $\{|m_{\ell+1}\rangle\}$ of the enlarged block.

5.1 Toy Example: Transverse Field Ising Model

Consider

$$H = -J \sum_i \sigma_i^z \sigma_{i+1}^z - h \sum_i \sigma_i^x \quad (50)$$

5.1.1 Block RG

- diagonalize a two-site block
- keep lowest states
- build effective Hamiltonian

As we have seen earlier, this method does not work properly.

5.1.2 DMRG Approach

- grow the chain
- calculate the ground state energy
- optimize basis using density matrix

The ground state energy converges exponentially with M_S . For many 1D systems extremely high precision can be achieved. As a simple example illustrating the DMRG algorithm we consider the one-dimensional transverse field Ising model. The Hamiltonian for a chain of length L is

$$H = -J \sum_{i=1}^{L-1} \sigma_i^z \sigma_{i+1}^z - h \sum_{i=1}^L \sigma_i^x. \quad (51)$$

Here $\sigma_i^{x,z}$ are Pauli matrices acting on site i , J denotes the interaction strength, and h is the transverse field.

Superblock Structure

In the DMRG algorithm the system is divided into two parts ($1 \dots l$ and $l+1 \dots L$). Combining all contributions we obtain the superblock Hamiltonian

$$H = H_S \otimes \mathbb{1} + \quad (52)$$

$$+ \mathbb{1} \otimes H_E + \quad (53)$$

$$- J \sigma_{L,\text{edge}}^z \otimes \sigma_{R,\text{edge}}^z \quad (54)$$

$$(55)$$

In the DMRG algorithm the operators acting on the system and environment blocks are represented in their truncated bases, while the operators on the two central sites are treated exactly. Diagonalizing this Hamiltonian gives the superblock ground state, which is then used to construct the reduced density matrix and update the block bases. All operators have to be rotated into the new basis. This is called the **renormalization** of the operators:

$$\tilde{O} = U^\dagger O U \quad (56)$$

Where

$$\rho = U \rho_{\text{diag}} U^\dagger \quad (57)$$

After this one can add a new site with the following rule:

$$H_{S,\text{new}} = \tilde{H}_S \otimes \mathbb{1} - J \tilde{\sigma}_{L,\text{edge}}^z \otimes \sigma^z - \tilde{\mathbb{1}} \otimes h \sigma^x \quad (58)$$

$$H_{E,\text{new}} = \mathbb{1} \otimes \tilde{H}_E - J\sigma^z \otimes \tilde{\sigma}_{\text{R,edge}}^z - h\sigma^x \otimes \tilde{\mathbb{1}} \quad (59)$$

$$\sigma_{\text{L,edge,new}}^z = \tilde{\mathbb{1}} \otimes \sigma^z \quad (60)$$

$$\sigma_{\text{R,edge,new}}^z = \sigma^z \otimes \tilde{\mathbb{1}} \quad (61)$$

$$H_{\text{new}} = H_{S,\text{new}} \otimes \mathbb{1} + \quad (62)$$

$$+ \mathbb{1} \otimes H_{E,\text{new}} + \quad (63)$$

$$- J\sigma_{\text{L,edge,new}}^z \otimes \sigma_{\text{R,edge,new}}^z \quad (64)$$

$$(65)$$

6 Finite-System DMRG and the Sweeping Algorithm

The infinite-system algorithm allows us to grow the system up to a desired length L . However, during the early stages of the growth the environment block is only a crude approximation to the true environment of the system. As a result, the basis states selected by the density matrix may not be optimal for the final system. To improve the accuracy, White introduced the *finite-system algorithm*. The key idea is to keep the total system size fixed and to iteratively improve the block bases by repeatedly optimizing them in the presence of the full system.

6.1 Environment Blocks

Unlike in the infinite-system algorithm, the environment blocks are not generated by mirroring the system block. Instead, they are taken from previously computed blocks stored during earlier steps of the algorithm. For each block size we store

- the truncated block basis,
- the renormalized Hamiltonian,
- all operators needed to construct the Hamiltonian.

This allows us to construct the environment block of any required size.

6.2 Sweeping Procedure

Once the system size L has been reached, the algorithm proceeds by moving the boundary between system and environment through the chain. This process is called a *sweep*. During a left-to-right sweep

$$\ell = 1, 2, 3, \dots, L - 3 \tag{66}$$

the system block grows while the environment block shrinks. At each step

1. Construct the superblock

$$S_\ell \bullet \bullet E_{L-\ell-2}. \tag{67}$$

2. Diagonalize the superblock Hamiltonian to obtain the ground state $|\psi\rangle$.
3. Construct the reduced density matrix of the enlarged system block.
4. Keep the M dominant eigenvectors and define the new block $S_{\ell+1}$.

When the system block reaches the right edge, the direction is reversed. In a right-to-left sweep the environment block grows while the system block shrinks.

6.3 Iterative Improvement

Each sweep improves the representation of the ground state because both blocks are optimized in the presence of the full environment. The process is repeated until convergence is reached. Typically the energy or other observables are monitored and the algorithm is stopped when their changes between successive sweeps become negligible.

7 Why DMRG Works

DMRG works well because:

- Ground states of local 1D Hamiltonians have low entanglement.
- Only a small number of states contribute significantly.
- The density matrix identifies these important states.

7.1 Summary

- The density matrix provides optimal truncation of Hilbert space.
- DMRG improves drastically over Block RG.
- Infinite algorithm grows the system.
- Finite algorithm (sweeping) improves accuracy.
- DMRG is the most powerful method for 1D quantum systems.

References

- [1] U. Schollwöck, *The density-matrix renormalization group*, Rev. Mod. Phys. (2005).
<https://arxiv.org/abs/cond-mat/0409292>